

2 History of Stretching

Stretching has been and continues to be a controversial training technique. When did it begin? Is it actually beneficial? Often, I have heard individuals state that if stretching was important for performance enhancement, you would see lions, tigers, and cheetahs meeting in groups (wearing spandex and knee braces) for a stretching routine before going on a hunt. While our mammalian cousins are not quite that organized, we do see animals stretching after sleeping or lounging for an extended period of time. This ritualized behaviour is called pandiculation. Pandiculation involves a voluntary contraction of the muscles, followed by a slow stretch/elongation and then relaxation (often with a yawn) (see Figure 2.1). The description is somewhat similar to contract-relax proprioceptive neuromuscular facilitation (PNF) (without the yawn), which was reported to be developed by Herman Kabat, but perhaps he stole the idea from his pet dog or cat (if he had a pet?). One simply has to watch a pet dog or cat and see them stretching their fore and



Figure 2.1 Pandiculation: involuntary stretching of soft tissues, with most mammals associated with transitions between cyclic biological behaviours, especially the sleep-wake rhythm

hind limbs after napping. Animals stretch (pandiculate) all the time! Is it a reflex ritual that actually benefits performance (hunting prey or escaping from predators)? Lions and tigers are not very accessible or easy subjects to recruit and I do not know of any stretch training studies using domesticated pets. Thus, there is no evidence for or against the effectiveness of pandiculation for performance enhancement. Obviously, animals (other than the human animal) have not organized and categorized their stretching and warm-up activities as humans have. Neither do animals stretch for such long durations or intensely as many human athletes or trained individuals. Of course, they also do not play tennis for 3–4 hours like professional tennis players, crash into each other for more than an hour like American football, rugby, and ice hockey players, perform *multiple* double or triple rotations of their bodies with twists for over 1–3 minutes like gymnasts, dancers, figure skaters, trampolinists, and others. Other mammals can be far more athletic than humans, but they tend to perform their athletic pursuits as a prey or a predator over a brief period, sporadically and without hours of training per day over months and years. We as humans tend to be far more obsessive, and compulsive in our activities and likewise our training for those activities. There is a common assumption that animals never get injured. There is no evidence for that assumption. Racehorses for example, commonly sustain muscle strains and have a high incidence of tendinopathy (23). Dogs suffer strains (muscles) and sprains (ligaments) (18). So maybe if they had better warm-up and stretching routines, they would experience less injuries? It is also argued that animals and ourselves often sprint without any warm-up (e.g., we are late for a bus!) and many times either catch the prey, escape from the predator or actually catch the bus. Thus, the contrarians argue there is no need for a warm-up. The problem with this argument is there is no control group! Nobody has taken a pride of lions and systematically stretched them over time (experimental group) and compared their performance to a control group of lions that did not stretch or only pandiculated. Perhaps, the lion would be incrementally faster if it did do a systematic warm-up. While, there is no evidence available for wild predators and prey, there is a large body of human research available regarding warm-up-induced positive changes to athletic performance (14), oxidative energy metabolism, increased lactate clearance, increased blood flow due to muscle capillary vasodilation (7), post-activation potentiation, neural conduction velocity, enzymatic cycling, tissue compliance, and a myriad of other contributing physiological factors (4, 5). We will cover many of these topics in the subsequent chapters.

When did humans start stretching in an organized manner with the hope of enhancing performance or decreasing the chance for injury? Although in historical texts, stretching is not always precisely listed as an activity preceding training or competition, it might be logical to assume that military personnel or athletes would have indulged in some kind of “limbering up” exercises to warm the muscles and body and increase the compliance or decrease the stiffness of the muscle and tendons. The martial arts of many Asian countries

are well known for their emphasis on extreme range of motion (ROM) in order to perform high kicks, acrobatic, and escape manoeuvres (think of Bruce Lee's athletic ability). The martial arts of the Chinese, Japanese and Aleut peoples, as well as Mongolian wrestling are suggested to have originated in the prehistoric era (11). Chinese boxing has been traced to the Zhou dynasty (1122–255 BCE). Martial arts like Kung Fu were influenced by Indian martial arts that spread to China in the fifth to sixth century CE (11). With the physical contortions, commonly observed with wrestling and martial arts, it would seem likely that the combatants would have needed to work on their flexibility to improve their chances of succeeding.

We are all familiar with the impressive flexibility of experienced yoga practitioners as they move from one difficult posture to another. Asia and specifically the Indus-Sarasvati civilization in Northern India are credited for the origins of Yoga during what is termed the pre-classical yoga period approximately 5,000 years ago and perhaps even earlier. However, this early yoga practice concentrated on the mind and spirit with little to no emphasis on the physical. Breath control exercises – the precursor of yoga – were implemented in China around 2600 BCE. It is not known if stretching exercises were also included but an exercise chart (168 BCE) of breathing and postures was developed for Tao Yin activities in the early Han dynasty. These exercises or postures were purported to cure specific illnesses (16). Perhaps, this is where the idea that stretching decreases injuries first began. Around the fourth century, during what is classified as the post-classical Yoga (classical Yoga period defined by Patanjali's Yoga-Sûtras "eight limbed path" to Samadhi or enlightenment starting in the second century CE), a system of practices was created to rejuvenate the body, prolong life and embrace the physical body to achieve enlightenment. Tantra Yoga, was developed to cleanse the body and mind. The evolution of these physical-spiritual connections and body-centred practices moved towards the development of Hatha Yoga (Figure 2.2).



Figure 2.2 Yoga positions

Stretching is specifically mentioned as an important component of an exercise regimen to prevent illness by Hua Tuo (104–208 CE). Hua Tuo suggested mimicking animal movements such as walking like a bear, and stretching the neck like a bird (even 2,000 years ago people noticed that animals stretched or at least pandiculated) among other animal-like movements. He emphasized combinations of breathing, bending, stretching and an assortment of postures which he labelled as the “Frolics of the Five Animals” (16). Tai Chi may trace its beginnings to these frolicking exercises.

In the Western civilizations, the Greeks held festivals (Tailteann Games: circa 1,800 BCE) that involved stone throwing, jumping, spear throwing, wrestling, and other activities (16). During pharaonic Egyptian times, athletes also wrestled, boxed, swam, ran, and lifted heavy objects in competition. One can imagine, that these athletes had some kind of pre-competition preparation (warm-up) and that especially for sports like wrestling where limbs can be forced and placed in extreme positions, preparatory stretching would have taken place.

Were these early stretching exercises more static or dynamic in nature? First of all, how do we define static and dynamic stretching? Static stretching involves lengthening a muscle until either a stretch sensation or the point of discomfort is reached and then holding the muscle in a lengthened position for a prescribed period of time (1, 2, 3). Dynamic stretching involves the performance of a controlled movement through the ROM of the active joint(s) (2, 3). Both types of stretching have gone through periods of popularity and disfavour. For example, Hua Tuo’s stretching the neck like a bird exercise would likely have involved a slow dynamic component to reach the end of the ROM and then a static component to hold that position. Dynamic stretching was popular in Persia where warriors and wrestlers starting around the first century AD used implements shaped like bowling pins called meels (Figure 2.3). While the heavy meels weighed approximately 50 pounds and would have been used for strength and



Figure 2.3 Persian meels and Indian clubs

power enhancement, the 2-pound meels were swung in patterns around the shoulder and would have been excellent for a dynamic warm up of the muscles and increasing the ROM. The Persians introduced this form of exercise to the Indian subcontinent in the thirteenth century. The people of the India subcontinent called this activity Persian yoga (16). It is quite likely that similar movement variations with a variety of weapons (i.e., short and long swords) would have been practiced by medieval knights in preparation for combat and competitions. Incorporating the movement of moderate to heavy loads as part of the pre-activity preparation (warm-up) is incorporated today to induce post-activation potentiation enhancement (PAPE) (6). PAPE not only improves subsequent strength and power performance by the aforementioned factors (i.e., neural conduction velocity, muscle vasodilation, increased rate of energy metabolism, etc.) but would also induce myosin light chain phosphorylation, which increases the sensitivity of the myofibrillar proteins (e.g., myosin, actin, tropomyosin, troponin) to calcium thus enhancing force output (4, 5).

Generally, such light dynamic movement for flexibility, quickness and muscular endurance was in the purview of men getting ready for battles or tournaments. However, in the mid-nineteenth century, exercises based on Swedish (Ling's) gymnastics were introduced to women in Europe and North America. They involved the graceful moving of arms, legs, neck, and head. Although their primary purpose (training for family life) would not have been to increase flexibility, such movements would have maintained or enhanced movement around the exercised joints. These *exercise-liberated* women could now be more effective at reaching farther across to make the bed in the mornings, extending further to scrub the floor under the furniture or reach deeper in the cupboard for the pots and pans for making dinner (training for family life!). Quite the liberation! Men, for their part, would still incorporate low intensity or light dynamic movements for sport preparation or war.

The late nineteenth century saw the emergence of a number of new team sports such as ice hockey (March 3, 1875: Montreal, Quebec, Canada), baseball (1672–1700: England), basketball (1891 by James Naismith (Canadian) at Springfield College, Massachusetts, USA), North American football (November 6, 1869, Rutgers vs. Princeton University, USA), volleyball (1895 by William G. Morgan from Springfield College, Massachusetts, USA) and others. The typical competitive zeal of the human athlete would instil a need to find a perceived advantage even in these early days of these new sports. One of those advantages could be a proper warm-up to prepare the body for competition and part of that warm-up would include light dynamic movements for “limbering up”.

But more importantly was the defence of your country and during the World Wars, soldiers were systematically trained to ensure they were ready for heavy military action. The systematic training of soldiers was incorporated during the world wars with scientific investigations of optimal resistance training routines sought out by Colonel DeLorme of the United States armed forces (9, 10). Subsequently, in the late 1950s and published in 1961, a Canadian, Dr. William Orban developed the 5BX (5 Basic eXercises)

programme (22). Though, it was originally targeted at male military personnel (air force pilots) who might not have access to training equipment and thus could perform calisthenics in most any location to stay fit, it spread to the general population. Orban also developed the XBX which were ten basic exercises modified for women. One of the stated objectives was to “Keep the important muscles and joints of the body supple and flexible.” Some of the exercises were quite dynamic and ballistic such as the toe touching exercises which involved bobbing up and down by flexing at the hips to touch the toes and then bounce back to an erect standing position.

However, the appeal of dynamic ballistic stretching activities diminished when it was noted that dynamic stretching of the muscles activated reflexes such as the myotactic (i.e., stretch) reflex, which results in reflexive contractions of the actively stretched muscle (19, 20). If the goal of stretching was to increase ROM, then it was reasoned that dynamic activities that elicited reflex muscle contractions while elongating the muscles could result in injury. Two forces would be working against each other with muscle elongation from stretching opposing stretch reflex-induced contractions. Hence, during the mid-1960s and thereafter, static stretching replaced ballistic or dynamic stretching as the predominant activity within a pre-activity warm-up routine to increase ROM (26, 27). Static stretching was recommended, as the slow movement into the stretch position and maintenance of a static stretch over a prolonged period minimized the reflexive firing of the muscle spindles (primarily the nuclear chain and bag fibres) that were activated by higher rates of stretch (12, 17). Hence, the attenuation of reflex activity with prolonged static stretching would presumably result in a more relaxed muscle and theoretically allow greater muscle lengths to be achieved. Thus, for the next 30 or more years, static stretching was the predominant form of stretching for warm-ups and flexibility.

In the 1970s another stretching technique also became more popular: proprioceptive neuromuscular acilitation (PNF) stretching (8). PNF was developed around 1946 by Herman Kabat, a neurophysiologist. The techniques evolved over time, but one popular variation of PNF was the contract-relax-agonist-contract (CRAC) method. If you wanted to stretch the hamstrings you would **CONTRACT** the hip flexors (i.e., quadriceps) till you reached your maximum ROM. Then you would **RELAX** as your partner held that elongated position, which would be followed by **CONTRACTION** of the hamstrings (**AGONISTS**). These variations of PNF were purported to induce a number of inhibitory reflex mechanisms (e.g., depressed H-reflex, reciprocal inhibition, autogenic inhibition: 21, 25) that would relax the muscle allowing the individual to reach even greater increases in ROM than with static stretching (8). You could not go to any team sport in the 1970s without seeing athletes pairing up to passively stretch and provide resistance to their partner's contractions of elongated muscles. However, not many people questioned at the time, why a technique like PNF that supposedly inhibited excitatory reflexes would be used in a warm-up that should excite the system in preparation for high intensity activity.

In the late 1990s and early 2000s scientific reports began to appear indicating that static stretching rather than enhancing subsequent performance might actually impair performance (13, 24). As the evidence began to mount throughout the early 21st century, static stretching was replaced with dynamic stretching as the major flexibility component of the warm-up. Only recently, has the evidence for static stretch-induced performance impairments been suggested to lack some ecological validity (practical reality or real-life application). A position stand/review by the Canadian Society for Exercise Physiology (2) published in 2016 documented that many of the static stretching studies did not employ a prior aerobic-style warm-up, stretched the muscle(s) for durations much longer than are typically used, did not include any dynamic sport specific activities after stretching and conducted the testing within 3–4 minutes of the experimental protocol. Another study found that just the knowledge of the previously published stretching impairment studies (expectancy bias) could negatively affect the results (15). Thus, the state of stretching within a warm-up and for improving ROM is in a state of flux and confusion. The objective of this book is to alleviate some of that confusion by critically analyzing the 5Ws of stretching with an “H” thrown in for good measure. That is, 1) **What** are the effects and physiological mechanisms underlying different types of stretching? 2) **Why** should we stretch? 3) **When** should we stretch? 4) **Who** are the major pioneers, innovators and researchers in this area? 5) **Where** does the science of flexibility and stretching go next and 6) **How** should we stretch or use other techniques to increase ROM?

Summary

Many animals elongate (stretch) their muscles after a period of rest. Although, these animals have pandiculated for eons, it is not known whether there are any performance benefits. There is inferential evidence that humans have stretched for thousands of years with the advent of Yoga (~3000 BCE) and martial arts (~1000 BCE) in Asia. The Greeks and Egyptians probably emphasized dynamic actions and stretches prior to their athletic competitions (~2000 BCE). Static stretching became more popular after the World Wars, while PNF stretching was popularized to a greater degree in the 1970s. Both stretching styles remained predominant until the 1990s, when research began to appear indicating that static stretching could lead to performance impairments. Since that time dynamic stretching has made a resurgence. The most recent studies suggest that the move away from static stretching may have been premature and based on impractical study designs.

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